Applied Machine Learning

Imputation

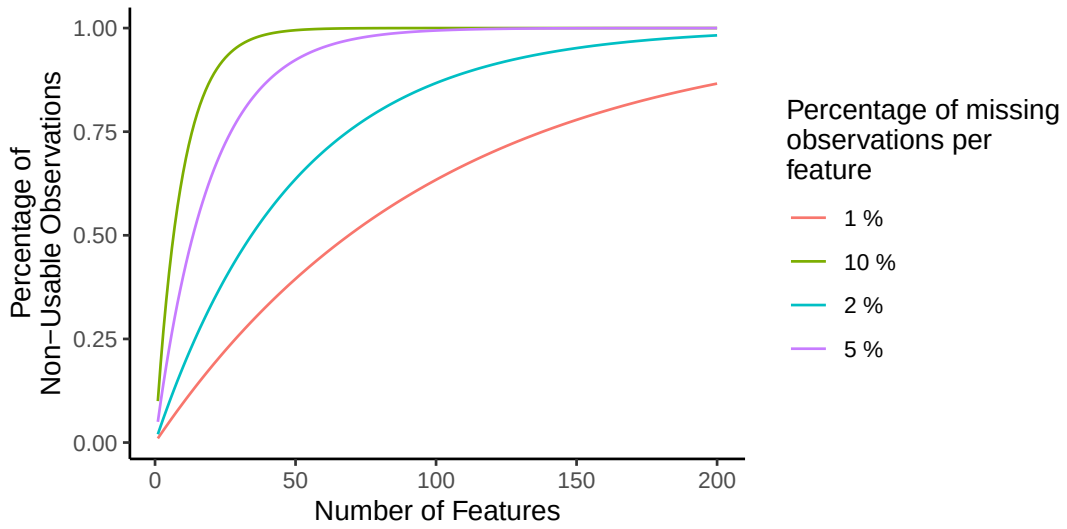
Learning goals

- Imputation

Imputation

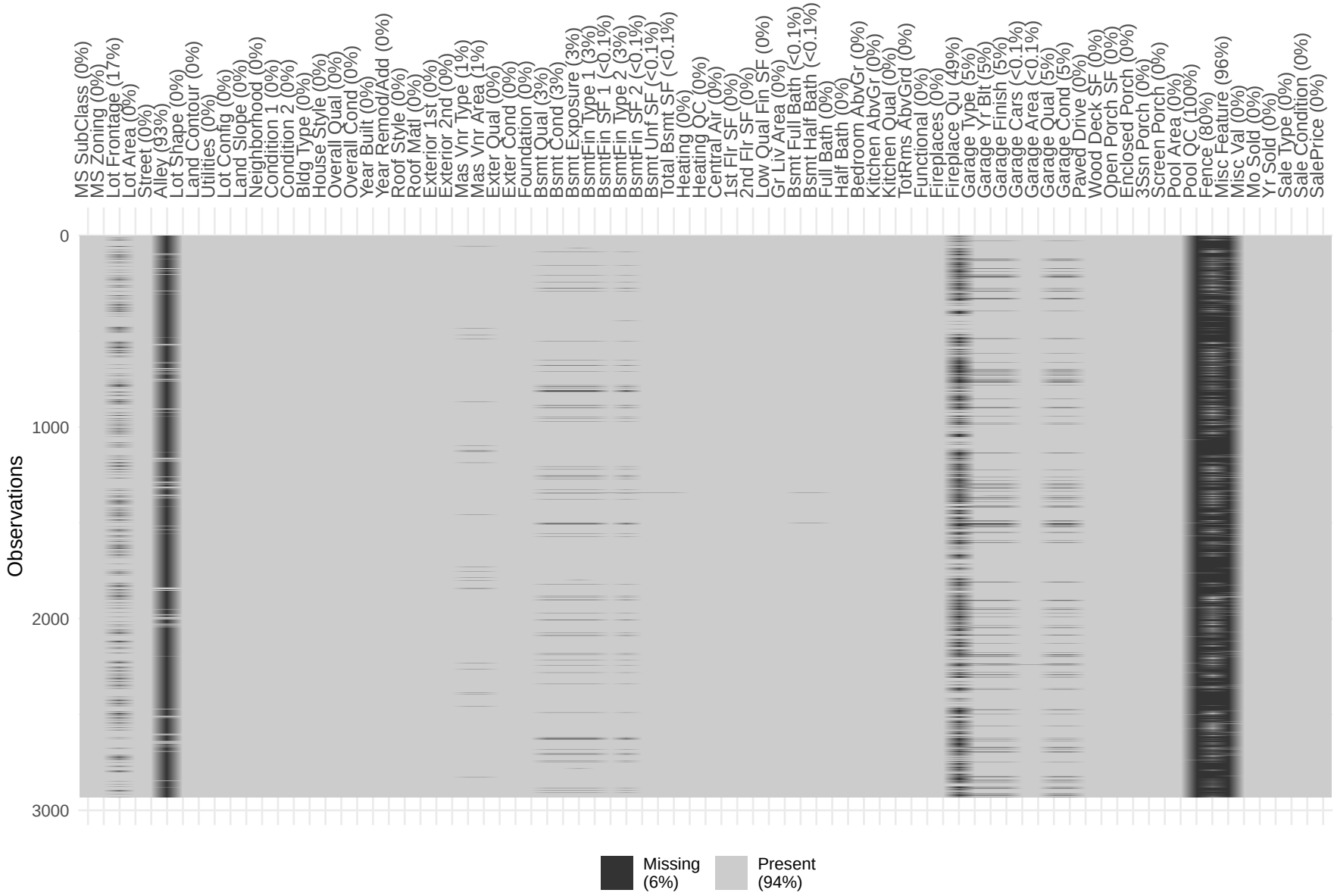
MOTIVATING EXAMPLE

- Assume each feature in your dataset has 2 % missing values.
- The missing values are randomly distributed over the observations.
- How many rows can be used if all observations that contain at least a missing value is dropped?



With 100 features and 2 % missing values only 13 % of our data can be used.

VISUALIZING MISSING VALUES



WHY IS DATA MISSING? REF

- faulty measurements
- unanswered questionnaire items
- unreported data

MISSINGNESS MECHANISMS

- MCAR (Missing Completely At Random)
  - Data is missing independently of missing or observed values
- MAR (Missing At Random)
  - Missingness depends only on observed values
- MNAR (Missing Not At Random)
  - Missingness depends on observed and missing values

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Which one is most realistic to assume in the real world?

MISSINGNESS MECHANISMS - EXAMPLES

- MCAR (Missing Completely At Random)
  - Participants skip a page of a questionaire because they flip two pages at once
- MAR (Missing At Random)
  - Certain medical examinations are conducted more often for certain patient groups (age groups, different gender)
  - Certain demographic groups are less likely to fill in surveys about depression (but demographic groups is often fully observable, e.g., gender)
- MNAR (Missing Not At Random)
  - Missingness depends on observed and missing values

POSSIBLE WAYS TO DEAL WITH MISSING VALUES

- Remove observations that contain missing values.
  But: Could lead to a very small dataset.
- Remove features that contain mostly missing values.
  But: Can lose (important) information.
- Use models that can handle missing values, e.g., (most) tree-based methods
  But: Restriction in model choice.
- **Imputation**
  $\rightarrow$ Replace missing values with *plausible* values.

REASONS TO IMPUTE MISSING VALUES

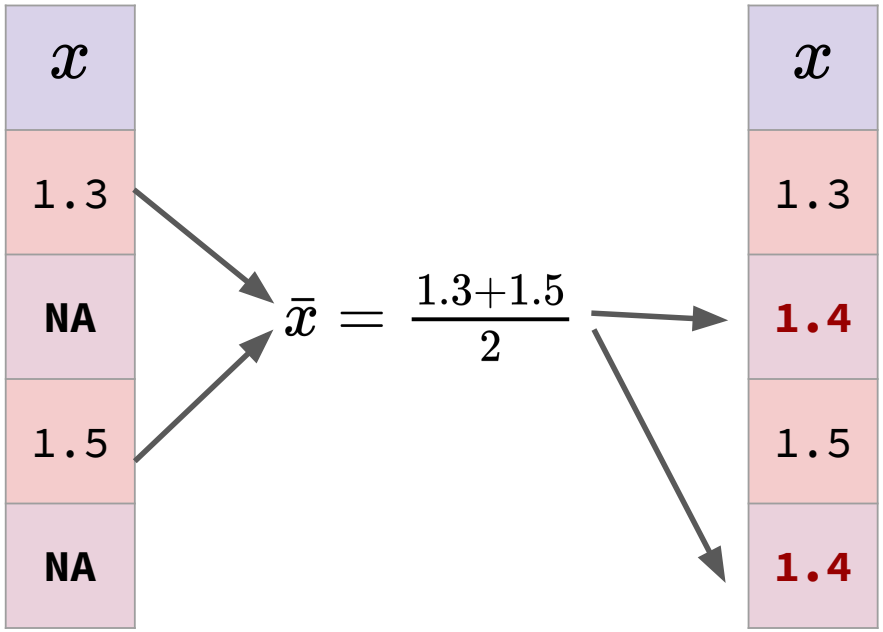
- Prediction
  - Goal: allow for maximal predictive performance
- Inference
  - Goal: estimate parameters such as mean and variance of the variable distribution
  - Not part of this lecture

$\rightarrow$

- Imputation depends on the goal
- Imputation needs to be benchmarked wrt the downstream task

SIMPLE IMPUTATION METHODS

A very simple imputation strategy is to replace missing values with univariate statistics, e.g. mean or median, of the feature:



SIMPLE IMPUTATION METHODS

The statistic used to impute the missing values has to match the type of the feature:

- Numeric features: mean, median, quantiles, mode, ...
- Categorical features: mode, ...

Alternatively missing values can be encoded with new values

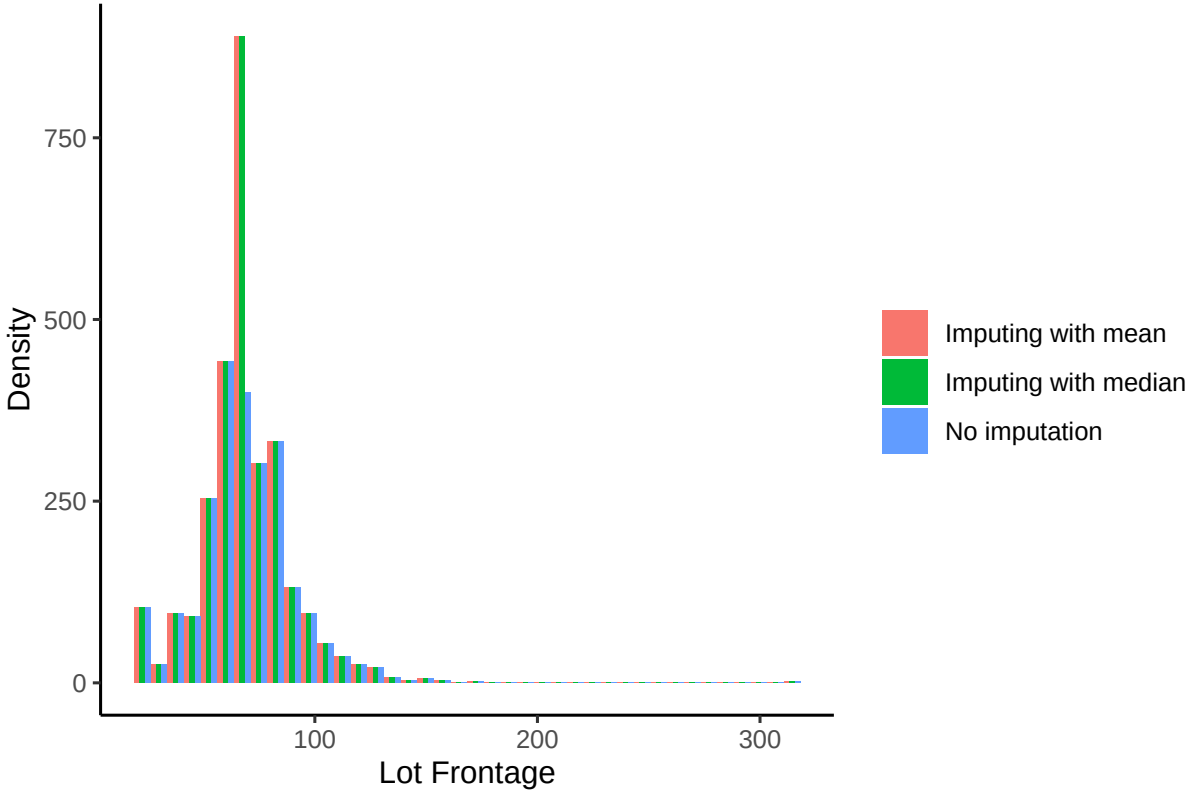
- Numeric features: `2*max`, ...
- Categorical features: `__MISS__`, ...

IMPUTATION NOTES

- To ensure that the information regarding which values were imputed is not lost, we can add a binary indicator variable.
  - Provides additional information in MNAR
- Domain knowledge is highly important: Missing Credit can mean that the individual has 0 debt.
- Encoding numeric values with out-of-range values has been shown to work well in practice for complex ML models.
  - This is especially useful for tree-based methods, as it allows separating observations with missing values in a feature.
  - But using out-of-range imputation when estimating global effects (e.g. in linear models) can skew the results

DISADVANTAGE OF CONSTANT IMPUTATION

By imputing a feature with one value we shift the distribution of that feature towards a single value.



IMPUTATION BY SAMPLING

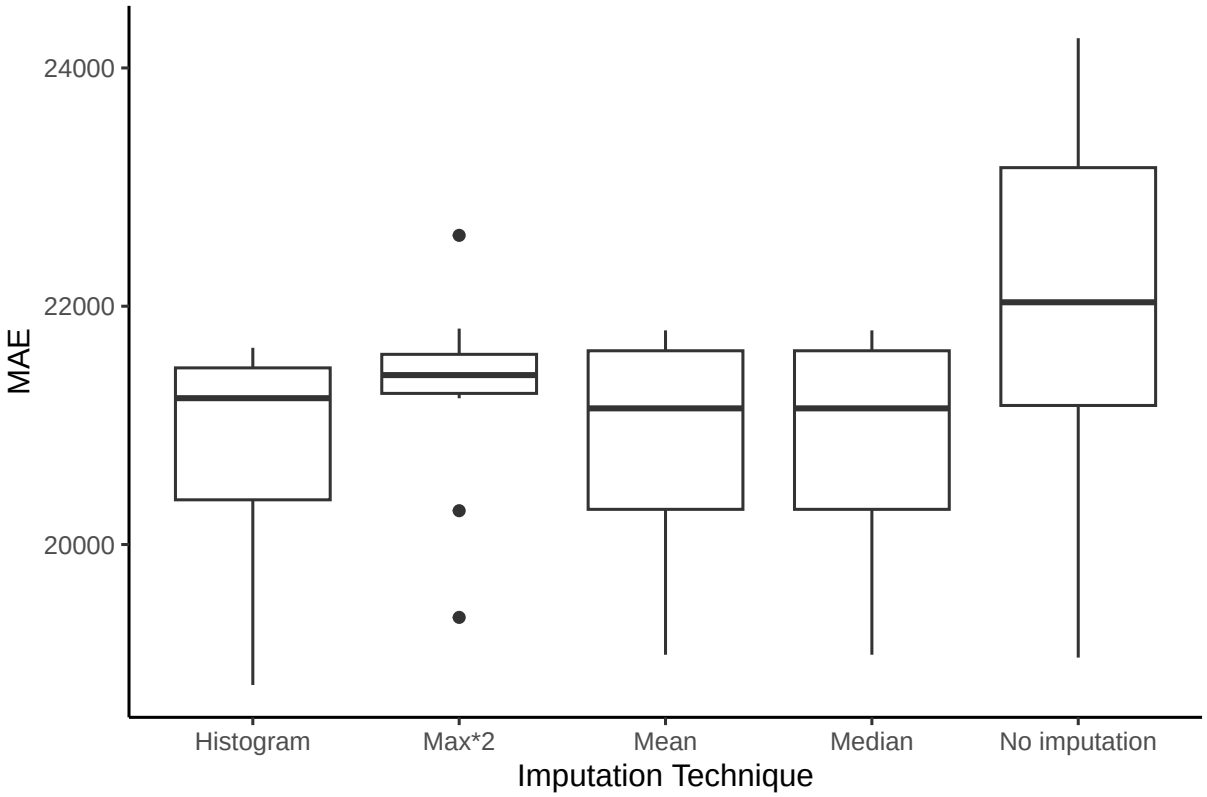
A way out of this problem is to sample values to replace each missing observation from

- the empirical distribution or histogram, for a numeric feature.
- the relative frequencies of levels, for a categorical feature.

This ensures that the distribution of the features does not change much.

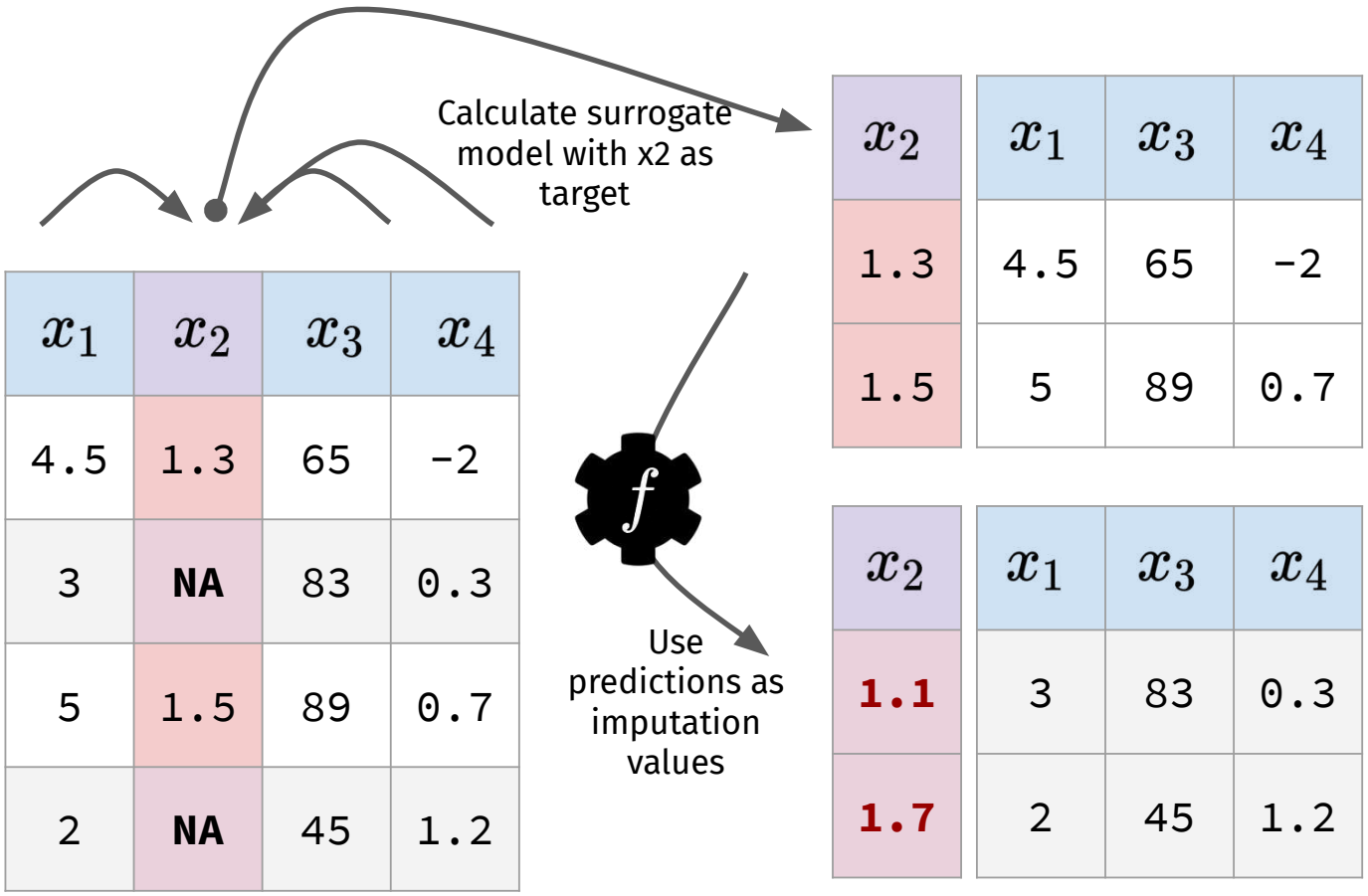
BENCHMARK OF SIMPLE IMPUTATION

To illustrate the effect of imputation on the performance we evaluate a linear model on the Ames housing dataset. Evaluation is done with a 10-fold cross-validation:



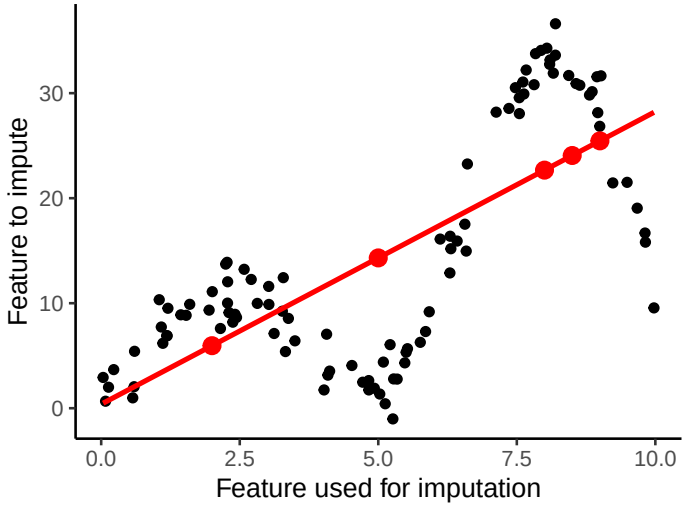
MODEL-BASED IMPUTATION

Instead of imputing a single value or sampling values it is desirable to take advantage of structure and correlation between features.



MODEL-BASED IMPUTATION: DRAWBACKS

- Choice of surrogate model has high influence on the imputation:



- Surrogate model should handle missing values itself, otherwise imputation *loop* may be necessary.
- Surrogate model hyperparameters can be tuned and can be different for each feature to impute.

A TALE OF FAILED IMPUTATION

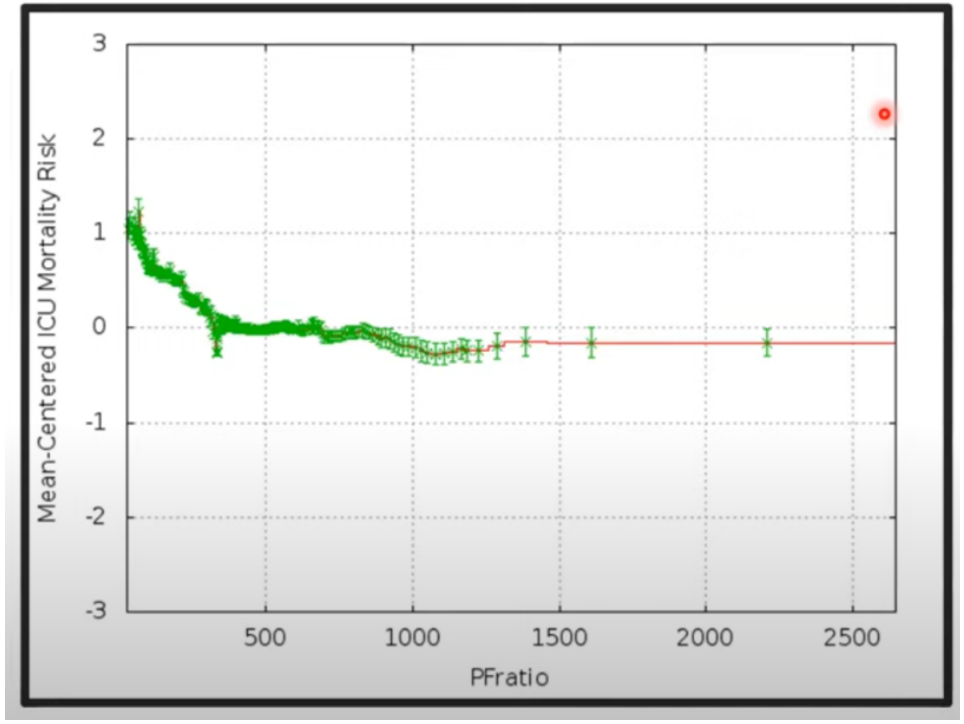
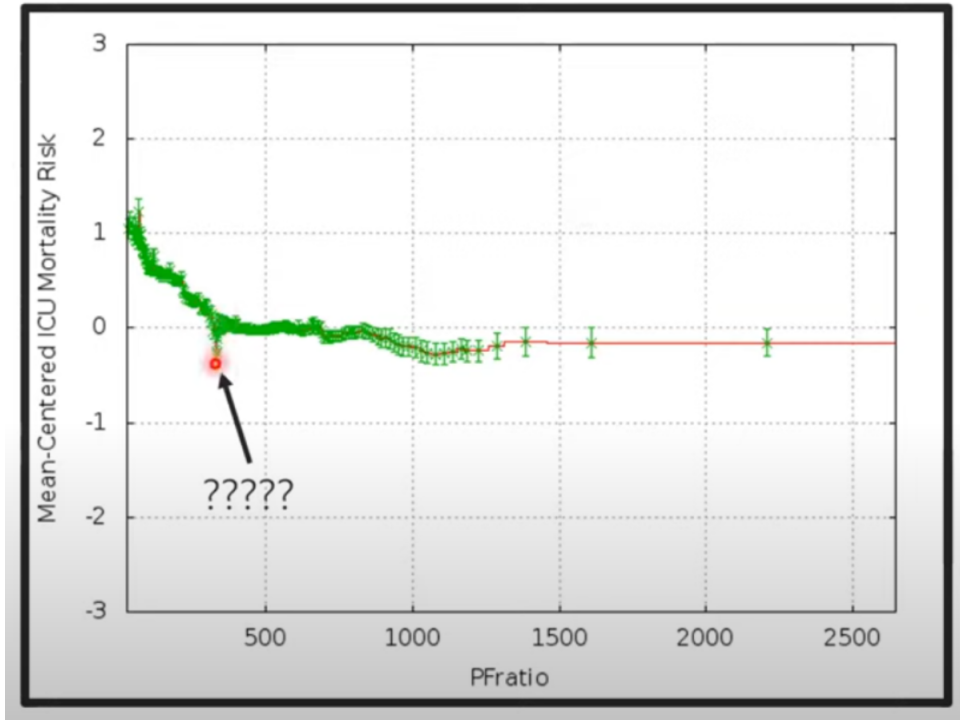


Figure: Figure from Rich Caruana's talk at the AutoML conference 2024, example starting at 43'31 REF

A TALE OF FAILED IMPUTATION 2



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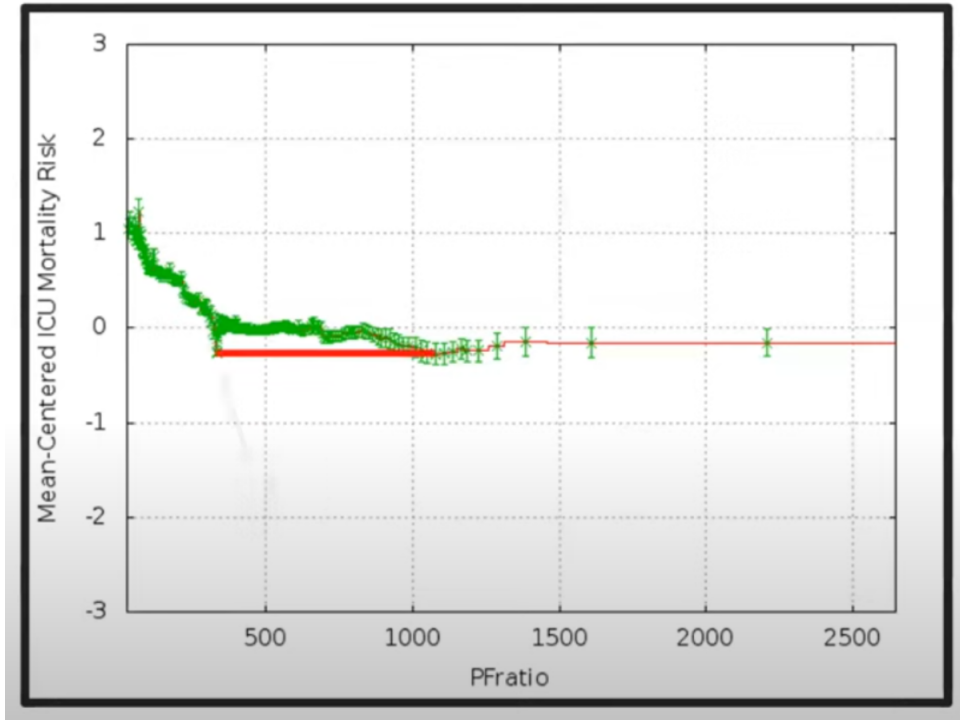
A TALE OF FAILED IMPUTATION 3



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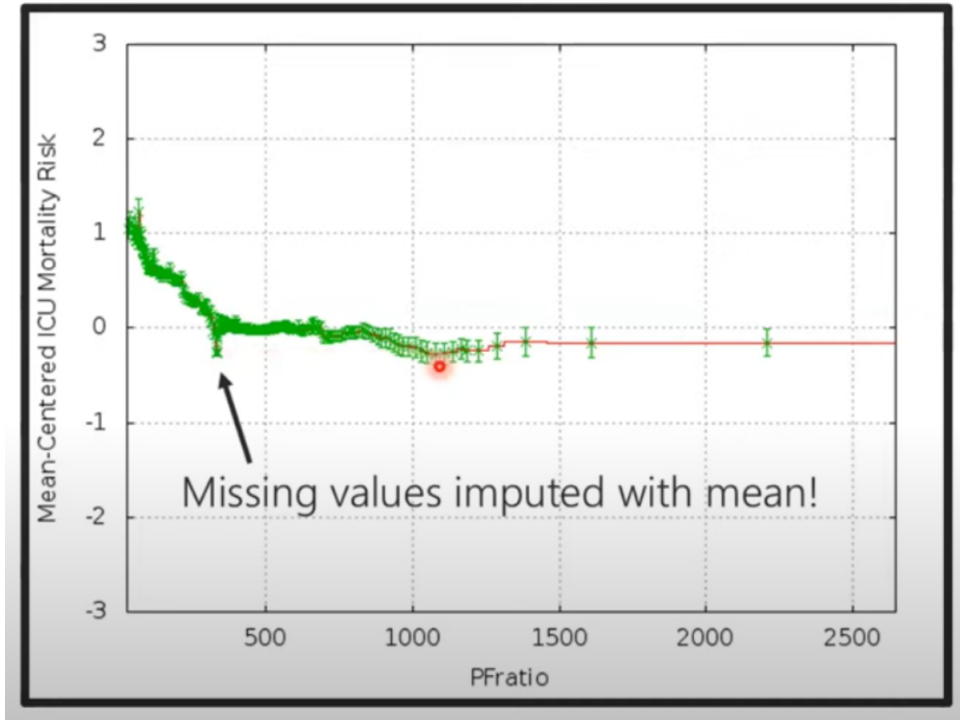
A TALE OF FAILED IMPUTATION 4



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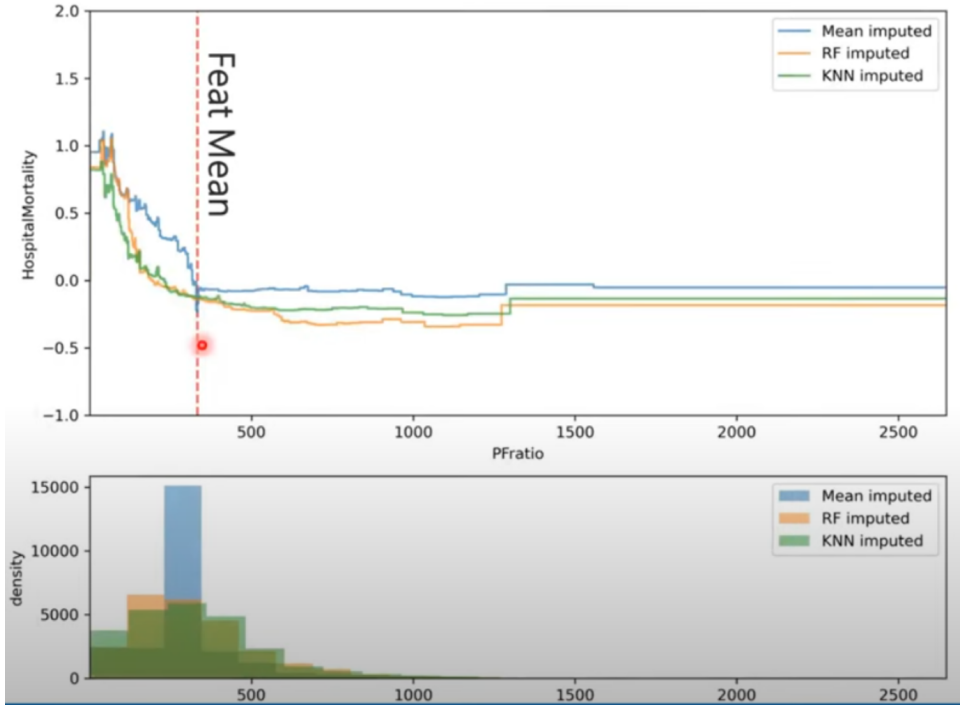
A TALE OF FAILED IMPUTATION 5



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TESTS FOR MISSINGNESS MECHANISMS

MAR vs MCAR

- Little's test
- Logistic regression for missingness
  1. For each column, create a new classification task: predict if the value is missing or not
  2. Use the remaining columns as features
  3. Fit logistic regression
  4. Examine coefficients and p-values

MNAR: no test available

CENSORED DATA: SPECIAL MISSINGNESS MECHANISMS

- Left censoring – a data point is below a certain value but it is unknown by how much.
- Interval censoring – a data point is somewhere on an interval between two values.
- Right censoring – a data point is above a certain value but it is unknown by how much.
- Type I censoring occurs if an experiment has a set number of subjects or items and stops the experiment at a predetermined time, at which point any subjects remaining are right-censored.
- Type II censoring occurs if an experiment has a set number of subjects or items and stops the experiment when a predetermined number are observed to have failed; the remaining subjects are then right-censored.
- Random (or non-informative) censoring is when each subject has a censoring time that is statistically independent of their failure time. The observed value is the minimum of the censoring and failure times; subjects whose failure time is greater than their censoring time are right-censored.

TAKEAWAYS OF A RECENT BENCHMARK

Imputation for prediction: beware of diminishing return. Le Morvan and Varoquaux, ICLR 2025 REF

- Better imputation performance does not result in better prediction performance for MNAR (w/o missingness indicator)
- Better imputation might be irrelevant b/c information is available in other features, unimportant, even with imputation it might be hard to learn a good downstream model
- More expressive models have less (relative) benefit from imputation
- Best on average: missForest + XGBoost + masking
- Open Questions:
  - performance of random draws
  - performance of multiple imputation
  - impact of performance on explainability and calibration
  - performance of models that can learn directly with missing value (advanced transformer architectures)

TAKEAWAYS

- There exist different reasons for missing data
- There exist different missingness mechanisms
- Different downstream tasks ask for different imputation strategies
- Different data modalities ask for different imputation strategies
- Understanding the data generating process helps deciding on the imputation strategy
- Common imputation strategies:
  - Constant value: mean, median
  - Imputation by sampling
  - Missingness indicator
  - Model-based

  $\rightarrow$ imputation selection is a CASH problem